# Meng Gao

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## Professional Experience

7/2019 to present Assistant Professor, Department of Geography, Hong Kong Baptist University, Kowloon, HK, China
8/2016-6/2019 Postdoctoral Fellow, Environmental Science and Engineering, Harvard University, Cambridge, MA, USA
1/2016-7/2016 Postdoctoral Fellow, Chemical Engineering, University of Iowa, Iowa City, IA, USA

## Education

12/2015 Ph.D. in Chemical Engineering, University of Iowa, Iowa City, IA, USA
5/2015 M.Eng. in Chemical Engineering, University of Iowa, Iowa City, IA, USA
6/2012 B.S. in Atmospheric Physics, Nanjing University of Information Science & Technology, Nanjing, China

## Selected Awards

2021 President's Award for Outstanding Performance as Young Researcher, HKBU
2021 Forbes 30 Under 30 Asia Class of 2021, Healthcare & Science
2020 Forbes China 30 Under 30 List, Science & Medical Health
2020 Atmospheric Sciences Division Outstanding Early Career Scientist Award, European Geosciences Union
2017 Atmospheric Chemistry Colloquium for Emerging Senior Scientists (ACCESS XIV), Department of Energy, USA (25 globally and biennially)
2016 Karl Kammermeyer Outstanding Research Award, University of Iowa, USA
2015 Graduate College Post-Comprehensive Research Award, University of Iowa, USA
2012 Karl Kammermeyer Fellowship, University of Iowa, USA

## Membership on Committees

2020-2022 World Meteorological Organization, Global Air Quality Forecasting and Information System (GAFIS) Steering Committee (1/15)
2021-now International Global Atmospheric Chemistry, Tropospheric Ozone Assessment Report, Phase II, East Asia Focus Working Group
2021-now Leader, Air Quality and Climate Change Working Group, Model Inter-Comparison Study for Asia Phase IV (MICS-Asia IV)

## Grants as Principal Investigator

- Assimilating surface $PM_{2.5}$ and ozone measurements to improve health exposure assessment and air quality forecasting in South China, UGC Early Career Scheme (ECS) (22201820), 2021-2023, 579,972 HKD
- 極軌和靜止衛星污染資料同化對我國大氣復合污染模擬預測的影響研究，NSFC-Young Scientists Fund, 2021-2023, 240,000 CNY
- Study on the Application of WRF-Chem over Hong Kong for Air Quality Simulation, Hong Kong Environmental Protection Department, 2021-2022, 1,024,000 HKD
- 排放與氣候因素對近十多年廣東冬季 $PM_{2.5}$ 濃度變化的影響， 廣東省自然科學基金-面上項目(2019A1515011633)，2019-2022, 100,000 CNY
- Co-occurrence and Exposure of Heat and Haze Hazards over China under a Warming Climate, Tier 2 Start-up Grant (SGT2) 2019/20, 499,000 HKD

- 排放與氣候因素對近十多年北京冬季 $PM_{2.5}$ 濃度變化的影響, 環境模擬與污染控制國家重點聯合實驗室（清華大學）開放課題(19K03ESPCT)，2019-2021, 80,000 CNY
- 中國與印度重污染區域臭氧的季節性與污染源特徵, 中國氣象局氣溶膠與雲降水重點開放實驗室開放課題 (KDW1901)，2019-2021, 30,000 CNY
- APEC 期間多污染物減排對黑碳光吸收影響的模擬研究，江蘇省大氣環境監測與污染控制高技術研究重點實驗室開放基金(KHK1902)，2019-2021, 50,000 CNY

## Publications (✉:Correspondence; #: Equal contribution; underscore: student/postdoc)

### Book & Book Chapter

- **Gao, M.**✉, Wang, Z. and Carmichael, G.R. eds., 2021. Air Pollution, Climate, and Health 1st Edition: An Integrated Perspective on Their Interactions, Elsevier.
- **Gao, M.**✉, Carmichael, G., Wang, Y., Saide, P., Liu, Z., Xin, J., Shan, Y. and Wang., Z.: "Chemical and Meteorological Feedbacks in the Formation of Intense Haze Events" in Air Pollution in Eastern Asia: An Integrated Perspective (ISSI Scientific Report Series), Springer, 1st ed. 2017 edition.

### Peer-reviewed Journal Publications

**Submitted (7)**

- Zhu, J.✉, Penner, J.✉, Garnier, A., Boucher, O., **Gao, M.**, Song, L., Deng, J., Liu, C., and Fu, P.: Climate effect on cirrus clouds due to decreased aviation during the COVID-19 pandemic, submitted to Science Advances.
- Lu, X., Ye, X., Zhou, M., Zhao, Y., Weng, H., Kong, H., Li, K., **Gao, M.**, Zheng, B., Lin, J., Zhou, F., Zhang, Q., Zhang, L.✉ and Zhang, Y.✉: The underappreciated role of agricultural soil nitrogen oxides emissions in ozone pollution regulation in North China, submitted to Nature Communications.
- Lin, L.#, **Gao, M.**#, Xiao, X.#, Knibbs, L.D., Morawska, L., Dharmage, S., Heinrich, J., Jalaudin, B., Lin, S., Guo, Y., Xu, S., Wu, Q., Chen, G., Yang, B., Zeng, X., Yu, Y., Hu, L. and Dong G.✉: Ultrafine particles, blood pressure and adult hypertension: a population-based survey in Northeast China, submitted to Environmental Research Letters.
- Xiao, X., **Gao, M.**, Knibbs, L., Heinrich, J., Dharmage, S., Moraswka, L., Lin, S. and Dong, G.✉: Is Greener the better? Associations between greenness and birth outcomes in both urban and non-urban settings, submitted to Environmental International.
- Li, B., Liu, C.✉, Hu, Q., Sun, M.✉, Zhang, C., Zhang, S., Zhu, Y., Liu, T., Guo, Y., Schultz, M.G., Carmichael, G. R. and **Gao., M.**✉: Deep learning breaks through the limit of satellite to monitor $PM_{2.5}$ in China, submitted to Proceedings of the National Academy of Sciences of the United States of America.
- **Gao, M.**✉, Yang, Y., Liao, H., Zhu, B., Zhang, Y., Liu, Z., Lu, X., Wang, C., Zhou, Q., Wang, Y., Zhang, Q., Carmichael, G. R., and Hu, J.✉: Reduced light absorption of black carbon (BC) and its influence on BC-boundary-layer interactions during APEC Blue, Atmos. Chem. Phys. Discuss. [preprint], https://doi.org/10.5194/acp-2021-170, in review, 2021.
- Yin, H., Liu, C.✉, Hu, Q., Liu, T., Wang, S., **Gao, M.**, Xu, S., Zhang, C., and Su, W.: Opposite impact of emission reduction during the COVID-19 lockdown period on the surface concentrations of $PM_{2.5}$ and $O_3$ in Wuhan, China, submitted to Environmental Pollution.
- Nitrogen isotope characteristics and source apportionment of atmospheric ammonium in urban cities during a haze process in Northern China Plan
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**2021 (12)**

- Liu, C., **Gao, M.**✉, Hu, Q., Brasseur, G. P. and Carmichael, G.R.: Stereoscopic monitoring: a promising strategy to advance diagnostic and prediction of air pollution, Bulletin of the American Meteorological Society, 102(4), E730-E737, 2021. (*IF=9.384*)

- **Gao, M.**✉, Zhou, Q., Yang, X., Li, Q., Zhang, S., Yung, K.K.L. and Guo, Y.✉: Nonlinear modulation of COVID-19 transmission by climate conditions. Meteorological Applications, 28(2), p.e1985, 2021. (*IF=1.685*)
- Sherman, P., **Gao, M.**, Song, S., Archibald, A. T., Abraham, N. L., Lamarque, J.-F., Shindell, D., Faluvegi, G., and McElroy, M. B.✉: Sensitivity of modeled Indian monsoon to Chinese and Indian aerosol emissions, Atmos. Chem. Phys., 21, 3593–3605, https://doi.org/10.5194/acp-21-3593-2021, 2021. (*IF=5.414*)
- Zong, L., Yang, Y.✉, **Gao, M.**, Wang, H., Wang, P., Zhang, H., Wang, L., Ning, G., Liu, C., Li, Y., and Gao, Z.: Synoptic drivers of co-occurring summertime ozone and $PM_{2.5}$ pollution in eastern China, Atmos. Chem. Phys., https://doi.org/10.5194/acp-2020-596, in press, 2021. (*IF=5.414*)
- Xiao, X., Wang, R., Knibbs, L., Jalaludin, B., Heinrich, J., Markevych, I., **Gao, M.**, Xu, S., Wu, Q., Zeng, X., Chen, G., Hu, L., Yang, B., Yu, Y. and Dong, G.✉: Street View Greenness is Associated with Lower Risk of Obesity in Adults: Findings from the 33 Chinese Community Health Study Street View, Environmental Research, in press, 2021. (*IF=5.026*)
- Zhu, Y., Hu, Q.✉, **Gao, M.**, Zhao, C., Zhang, C., Liu, T., Tian, Y., Yan, L., Su, W., Hong, X. and Liu, C.: Quantifying contributions of local emissions and regional transport to NOX in Beijing using TROPOMI constrained WRF-Chem simulation, Remote Sensing, in press, 2021. (*IF=4.509*)
- Zhang, Y., Zhao, Y.✉, **Gao, M.**, Bo, X. and Nielsen, C.: Air quality and health benefits from ultra-low emission control policy indicated by continuous emission monitoring: A case study in the Yangtze River Delta region, China, Atmospheric Chemistry and Physics, in press, 2021. (*IF=5.414*)
- Yang, B., Qu, Y., Guo, Y., Markevych, I., Heinrich, J., Bloom, M. S., Bai, Z., Knibbs, L. C., Li, S., Chen, G., Jalaludin, B., Morawska, L, **Gao, M.**, Han, B., Yu, Y., Liu, X., Ou, Y., Mai, J., Gao, X., Wu, Y., Nie, Z., Zeng, X., Hu, L., Lin, S., Liu, X. and Dong, G.✉: Maternal exposure to ambient air pollution and congenital heart defects in China, Environmental International, 153, 106548, 2021. (*IF=7.577*)
- Wang, P., Shen, J., Zhu, S., **Gao, M.**, Ma, J., Liu, J., Gao, J. and Zhang H.✉: The aggravated short-term $PM_{2.5}$-related health risk due to atmospheric transport in the Yangtze River Delta, Environmental Pollution, 275, p.116672., 2021. (*IF=6.792*)
- Park, R. J., Oak, Y. J., Emmons, L. K., Kim, C. H., Pfister, G. G., Carmichael, G. R., Saide, P.E., Cho, S.Y., Kim, S., Woo, Crawford, J.H., Gaubert, B., Lee, H.J., Park, S.Y., Jo, Y.J., **Gao, M.**, Tang, B., Stanier, C.O., Shin, S.S., Park, H.Y., Bae, C. and Kim, E.: Multi-model intercomparisons of air quality simulations for the KORUS-AQ campaign. Elementa: Science of the Anthropocene, 9(1)., 2021. (*IF=4.212*)
- Song, S.✉, Ma, T., Zhang, Y., Shen, L., Liu, P., Li, K., Zhai, S., Zheng, H., **Gao, M.**, Moch, J. M., Duan, F., He, K., and McElroy, M. B.: Global modeling of heterogeneous hydroxymethanesulfonate chemistry, Atmos. Chem. Phys., 21, 457–481, https://doi.org/10.5194/acp-21-457-2021, 2021. (*IF=5.414*)
- Huang, W., Yu, Y.✉, Yin, Z., Chen, H. and **Gao, M.**: Appreciable role of stratospheric polar vortex in the abnormal diffusion of air pollutant in North China and implications for prediction, Atmospheric Environment, in press, 2021. (*IF=4.012*)

**2020 (23)**

- **Gao, M.**✉, Liu, Z., Zheng, B., Ji, D., Sherman, P., Song, S., Xin, J., Liu, C., Wang, Y., Zhang, Q., Wang, Z.✉, Carmichael, G., and McElroy, M.✉: China's emission control measures have suppressed unfavorable influences of climate on wintertime $PM_{2.5}$ concentrations in Beijing since 2002, Atmospheric Chemistry and Physics, 20, 1497–1505, https://doi.org/10.5194/acp-20-1497-2020, 2020. (*IF=5.414*)
- **Gao, M.**✉, Han, Z.✉, Tao, Z., Li, J., Kang, J.-E., Huang, K., Dong, X., Zhuang, B., Li, S., Ge, B., Wu, Q., Lee, H.-J., Kim, C.-H., Fu, J. S., Wang, T., Chin, M., Li, M., Woo, J.-H., Zhang, Q.,

Cheng, Y., Wang, Z., and Carmichael, G. R.✉: Air Quality and Climate Change, Topic 3 of the Model Inter-Comparison Study for Asia Phase III (MICS-Asia III), Part II: aerosol radiative effects and aerosol feedbacks. Atmospheric Chemistry and Physics, 20, 1147–1161, https://doi.org/10.5194/acp-20-1147-2020, 2020. (*IF=5.414*)

- **Gao, M.✉,** Gao, J., Zhu, B., Kumar, R., Lu, X., Song, S., Zhang, Y., Jia, B., Wang, P., Beig, G., Hu, J., Ying, Q., Zhang, H., Sherman, P., and McElroy, M. B.✉: Ozone Pollution over China and India: Seasonality and Sources, Atmospheric Chemistry and Physics, 20, 4399–4414, https://doi.org/10.5194/acp-20-4399-2020, 2020. (*IF=5.414*)
- Jia, B., Wang, Y.✉, Wang, C., Zhang, Q., **Gao, M.✉** and Yung, K.L.L: Sensitivity of $PM_{2.5}$ to $NO_x$ emissions and meteorology in North China based on observations. Science of the Total Environment, p.142275, 2020. (*IF=6.551*)
- Jia, B., **Gao, M.✉**, Zhang, X., Xiao, X., Zhang, S. and Yung, K.K.L.✉. Rapid increase in mortality attributable to PM2. 5 exposure in India over 1998-2015. Chemosphere, p.128715, 2020. (*IF=5.778*)
- Liao, Z., Z. Ling, **M. Gao**, J. Sun, W., Zhao, P. Ma, J. Quan✉ and Fan S.✉, Tropospheric ozone variability over Hong Kong based on recent 20-year (2000–2019) Ozonesonde observation, Journal of Geophysical Research: Atmospheres, 126(3), p.e2020JD033054, 2020. (*IF=3.820*)
- Saide, P. E.✉, **Gao, M.**, Lu, Z., Goldberg, D., Streets, D. G., Woo, J.-H., Beyersdorf, A., Corr, C. A., Thornhill, K. L., Anderson, B., Hair, J. W., Nehrir, A. R., Diskin, G. S., Jimenez, J. L., Nault, B. A., Campuzano-Jost, P., Dibb, J., Heim, E., Lamb, K. D., Schwarz, J. P., Perring, A. E., Kim, J., Choi, M., Holben, B., Pfister, G., Hodzic, A., Carmichael, G. R., Emmons, L., and Crawford, J. H.: Understanding and improving model representation of aerosol optical properties for a Chinese haze event measured during KORUS-AQ, Atmospheric Chemistry and Physics, https://doi.org/10.5194/acp-2019-1022, in press, 2020. (*IF=5.414*)
- Yang, B.Y., Hu, L.W., Jalaludin, B., Knibbs, L.D., Markevych, I., Heinrich, J., Bloom, M.S., Morawska, L., Lin, S., Jalava, P., Roponen, M., **Gao, M.**, Chen, D.H., Zhou, Y., Yu, H., Liu, R., Zeng, X., Zeeshan, M., Guo, Y., Yu, Y. and Dong, G.✉: Association Between Residential Greenness, Cardiometabolic Disorders, and Cardiovascular Disease Among Adults in China. JAMA Network Open, 3(9), pp.e2017507-e2017507, 2020. (*IF=5.032*)
- Yu, H., Guo, Y., Zeng, X., **Gao, M.**, Yang, B.Y., Hu, L.W., Yu, Y. and Dong, G.H.✉: Modification of caesarean section on the associations between air pollution and childhood asthma in seven Chinese cities. Environmental Pollution, 267, p.115443, 2020. (*IF=6.792*)
- Wang, Y., Wild, O., Chen, X., Wu, Q., **Gao, M.**, Chen, H., Qi, Y. and Wang, Z.✉: Health impacts of long-term ozone exposure in China over 2013–2017. Environment International, 144, p.106030, 2020. (*IF=7.577*)
- Liu, L.✉, Zhang, X.✉, Xu, W., Liu, X., Li, Y., Wei, J., **Gao, M.**, Bi, J., Lu, X., Wang, Z. and Wu, X.: Challenges for global sustainable nitrogen management in agricultural systems. Journal of Agricultural and Food Chemistry, 68(11), pp.3354-3361., 2020. (*IF=4.192*)
- Wang, H., Li, J., **Gao, M.**, Chan, T.C., Gao, Z., Zhang, M., Li, Y., Gu, Y., Chen, A., Yang, Y.✉, and Ho, H.C.✉: Spatiotemporal variability in long-term population exposure to PM2.5 and lung cancer mortality attributable to PM2.5 across the Yangtze River Delta (YRD) region over 2010 - 2016: A multistage approach. Chemosphere, 10.1016/j.chemosphere.2020.127153, 2020. (*IF=5.778*)
- Liu, K., Yang, B.Y., Guo, Y., Bloom, M.S., Dharmage, S.C., Knibbs, L.D., Heinrich, J., Leskinen, A., Lin, S., Morawska, L., Jalaludin, B., Markevych, I., Jalava, P., Komppula, M., Yu, Y., **Gao, M.**, Zhou, Y., Yu, H.Y., Hu, L.W., Zeng, X.W. and Dong, G.H.✉: The role of influenza vaccination in mitigating the adverse impact of ambient air pollution on lung function in children: new insights from the Seven Northeastern Cities Study in China. Environmental Research, p.109624, 2020. (*IF=5.026*)
- Lu, X., Zhang, L.✉, Wang, X., **Gao, M.**, Li, K., Zhang, Y., Yue, X. and Zhang, Y.✉: Rapid increases in warm-season surface ozone and resulting health impact over China since 2013. Environmental Science & Technology Letters, 7, 4, 240-247, 2020. (*IF=7.678*)
- Shen, L., Hao, F., **Gao, M.**, Wang, H.✉, Zhu, B., Gao, J., Cheng, Y. and Xie, F.: Real-time

geochemistry of urban aerosol during a heavy dust episode by single-particle aerosol mass spectrometer: Spatio-temporal variability, mixing state and spectral distribution. Particuology, https://doi.org/10.1016/j.partic.2020.02.001, 2020. (*IF=2.787*)

- Wang, Y., Wild, O., Chen, H., **Gao, M.**, Wu, Q., Chen, X., and Wang, Z.✉: Acute and chronic health impacts of PM2.5 in China and the influence of interannual meteorological variability, Atmospheric Environment, 229, 117397, 2020. (*IF=4.012*)
- Sun, S., Jin, J., Xia, M., Liu, Y., **Gao, M.**, Zou, C., Wang, T., Liu, Y., Wu, L., Mao, H.✉, and Wang, P.✉: Vehicle emissions in a middle-sized city of China: current status and future trends, Environmental International, 137, 105514, 2020. (*IF=7.577*)
- Tao, Z.✉, Chin, M., **Gao, M.**, Kucsera, T., Kim, D., Bian, H., Kurokawa, J., Wang, Y., Carmichael, G. R., Wang, Z., and Akimoto, H.: Evaluation of NU-WRF Performance on Air Quality Simulation under Various Model Resolutions – An Investigation within Framework of MICS-Asia Phase III, Atmospheric Chemistry and Physics, 20, 2319–2339, https://doi.org/10.5194/acp-20-2319-2020, 2020. (*IF=5.414*)
- Zeng, Z., Wang, Z., Gui, K., Yan, X., **Gao, M.**, Luo, M., Geng, H., Liao, T., Li, X., An, J., Liu, H., He, C., Ning, G., Yang, Y.✉**:** Daily Global SolarRadiation in China Estimated from High-density Meteorological Observations: A Random Forest Model Framework, Earth and Space Science, 7(2), p.e2019EA001058., 2020. (*IF=2.313*)
- Fu, P., **Gao, M.** and Yung, K. K.L.✉: Association of intestinal disorders with Parkinson's disease and Alzheimer's disease: A systematic review and meta-analysis, ACS Chemical Neuroscience, 11, 3, 395-405, 2020. (*IF=4.486*)
- Xu, Y.[#,✉], Wu, X.[#], Kumar, R., Barth, M., Diao, C., **Gao, M.**, Lin, L., Jones, B., and Meehl, G. A.: Substantial increase in the joint occurrence and human exposure of Heatwave and High-PM Hazards (HHH) over South Asia in the mid-21st century, AGU Advances, 1, e2019AV000103.https://doi.org/10.1029/2019AV000103, 2020.
- Yang, B.Y., Liu, K.K., Markevych, I., Knibbs, L.D., Bloom, M.S., Dharmage, S.C., Lin, S., Morawska, L., Heinrich, J., Jalaludin, B., **Gao, M.**, Guo, Y., Zhou, Y., Huang, W., Yu, H., Zeng, X., Hu, L., Hu, Q., Dong, G.✉: Association between residential greenness and metabolic syndrome in Chinese adults. Environment International, 135, p.105388, 2020. (*IF=7.577*)
- Shan, Y., Wilcox, E.M.✉, Gao, L., Lin, L., Mitchell, D.L., Yin, Y., Zhao, T., Zhang, L., Shi, H. and **Gao, M.**: Evaluating Errors in Gamma Function Representations of the Rain Drop Size Distribution: A method for determining the optimal parameter set for use in bulk microphysics schemes. Journal of the Atmospheric Sciences, https://doi.org/10.1175/JAS-D-18-0259.1, 2020. (*IF=3.194*)

**2019 (12)**

- **Gao, M.**, Sherman, P., Song, S., Yu, Y., Wu, Z. and McElroy, M. B.✉: Seasonal Prediction of Indian Wintertime Aerosol Pollution using the Ocean Memory Effect. **Science Advances**, 5(7), eaav4157, 2019. (*IF=13.116*)
- Sherman, P., **Gao, M.**, Song, S., Ohiomoba, P., Archibald, A., McElroy, M.✉: The Influence of Dynamics and Emissions Changes on China's Wintertime Haze, Journal of Applied Meteorology and Climatology, https://doi.org/10.1175/JAMC-D-19-0035.1, 2019. (*IF=2.506*)
- Ji, D.✉, **Gao, M.**, Maenhaut, W.✉, He, J., Wu, C., Cheng, L., Gao, W., Sun, Y., Sun, J., Xin, J., Wang, L., Wang, Y.: The Carbonaceous Aerosol Levels still Remain A Challenge in the Beijing-Tianjin-Hebei region of China: insights from continuous high temporal resolution measurements in multiple cities. Environmental International, 126, 171-183, 2019. (*IF=7.577*)
- Song, S.✉, **Gao, M.**, Xu, W., Sun, Y.✉, Worsnop, D., Jayne, J., Zhang, Y., Zhu, L., Li, M., Zhou, Z., Cheng, C., Lv, Y., Wang, Y., Peng, W., Xu, X., Lin, N., Wang, Y., Wang, S., Munger, J. W., Jacob, D., McElroy, M.B.✉: Possible Heterogeneous Hydroxymethanesulfonate (HMS) Chemistry in Northern China Winter Haze and Implications for Rapid Sulfate Formation. Atmospheric Chemistry and Physics, 19, 1357-1371, 2019. (*IF=5.414*)
- Yue, S., Bikkina, S., **Gao, M.**, Barrie, L.A., Kawamura, K. and Fu, P.✉: Sources and Radiative Absorption of Water-Soluble Brown Carbon in the High Arctic Atmosphere. Geophysical

Research Letters, https://doi.org/10.1029/2019GL085318, 2019. (*IF=4.580*)

- Song, S., Nenes, A., **Gao, M.**, Zhang, Y., Liu, P., Shao, J., Ye, D., Xu, W., Liu, L., Sun, Y.✉, Liu, B.✉, Wang, S.✉ and McElroy, M.B.: Thermodynamic Modeling Suggests Declines in Water Uptake and Acidity of Inorganic Aerosols in Beijing Winter Haze Events during 2014/2015–2018/2019. Environmental Science & Technology Letters, 6(12), pp.752-760, 2019. (*IF=7.678*)
- Zhou, W., **Gao, M.,** He, Y., Wang, Q., Xie, C., Xu, W., Zhao, J., Du, W., Qiu,Y., Lei, L., Fu, P., Wang, Z., Worsnop, D.R., Zhang, Q., Sun, Y.✉: Response of Aerosol Chemistry to Clean Air Action in Beijing, China: Insights from two-year ACSM measurements and model simulations, Environmental Pollution, https://doi.org/10.1016/j.envpol.2019.113345, 2019. (*IF=6.792*)
- Cui, Y., Ji, D.✉, Chen, H., **Gao, M**., Maenhaut, W., He, J. and Wang, Y.: Characteristics and Sources of Hourly Trace Elements in Airborne Fine Particles in Urban Beijing, China. Journal of Geophysical Research: Atmospheres, https://doi.org/10.1029/2019JD030881, 2019. (*IF=3.820*)
- Liu, Z., Hu, B., Ji, D., Cheng, **M., Gao**, W., Shi, S., Xie, Y., Yang, S., Gao, M., Fu, H., Chen, J.✉ and Wang, Y.✉: Characteristics of Fine Particle Explosive Growth Events in Beijing, China: seasonal variation, chemical evolution pattern and formation mechanism. Science of The Total Environment, 687, 1073-1086, 2019. (*IF=6.551*)
- Goldberg, D.✉, Saide, P., Lamsal, L., de Foy, B., Lu, Z., Woo, J., Kim, Y., Kim, J., **Gao, M.**, Carmichael, G. R. and Streets, D.: A Top-down Assessment using OMI $NO_2$ Suggests an Underestimate in the NOx Emissions Inventory in Seoul, South Korea during KORUS-AQ. Atmospheric Chemistry and Physics, 19(3), 1801-1818, 2019. (*IF=5.414*)
- Zhou, M., Zhang, L.✉, Chen, D., Gu, Y., Fu, T., **Gao, M.**, Zhao, Y., Lu, X. and Zhao, B.: The Impact of Aerosol-Radiation Interactions on the Effectiveness of Emission Control Measures. Environmental Research Letters, 14, 024002, 2019. (*IF=6.192*)
- Lu, X., Zhang, L.✉, Zhao, Y., Jacob, D.J.✉, Hu, Y.✉, Hu, L., **Gao, M.**, Liu, X., Petropavlovskikh, I., McClure-Begley, A. and Querel, R.: Surface and Tropospheric Ozone Trends in the Southern Hemisphere since 1990: Possible Linkages to Poleward Expansion of the Hadley Circulation. Science Bulletin, 64(6), 400-409, 2019. (*IF=6.227*)

**2018 (11)**

- **Gao, M.**✉, Ding, Y., Song, S., Lu, X., Chen, X.✉ and McElroy, M. B.✉: Secular Changes in Wind Energy Potential in India Associated with Warming Indian Ocean. **Science Advances**, 4(12), p.eaat5256, 2018. (*IF=13.116*)
- **Gao, M.**✉, Beig, G., Song, S., Zhang, H., Hu, J., Ying, Q., Liang, F., Liu, Y., Wang, H., Lu, X., Zhu, T., Carmichael, G. R., Nielsen, C. P.✉ and McElroy, M. B.: The Impact of Power Generation Emissions on Ambient $PM_{2.5}$ Pollution and Human Health in China and India. Environmental International, 121, 250-259, 2018. (*IF=7.577*)
- **Gao, M.**✉, Ji, D.✉, Liang, F. and Liu, Y.: Attribution of Aerosol Direct Radiative Forcing in China and India to Emitting Sectors. Atmospheric Environment, 190, 35-42, 2018. (*IF=4.012*)
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- **Gao, M.**✉, Guttikunda, S. K., Carmichael, G. R., Wang, Y., Liu, Z., Stanier, C. O., Saide, P. E. and Yu, M.: Health Impacts and Economic Losses Assessment of the 2013 Severe Haze Event in Beijing Area. Science of the Total Environment, 511, 553-561, 2015. **(ESI Highly Cited Paper)** (*IF=6.551*)
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**Thesis**

- **Gao, M.**: Improving Understanding of Haze Pollution in the North China Plain via Atmospheric Modeling and Data Assimilation, University of Iowa, 2015

## Oral Presentations

- 6/3/2021, **Seminar on "Recovery from COVID-19"**: COVID-19, Air Quality, and Climate, Hong Kong
- 5/21/2021, **Transnational and Transdisciplinary Lessons from the Covid-19 Pandemic**: Distinct regimes of ozone response to COVID-19 lockdown in China, Hong Kong
- 4/19/2021, **EGU General Assembly 2021:** Aerosol-Weather-Climate Interactions in Highly Polluted Regions, Online.
- 4/16/2021, **蘭州大學大氣科學學院**: 污染與氣溶膠輻射反饋的相互作用研究, Online.
- 4/1/2021, **中國清潔空氣政策夥伴關係（CCAPP）學術沙龍**: 亚洲气溶胶-天气-气候多尺度相互作用机制研究, Online.
- 12/9/2020, **中國環境科學學會大氣環境分會**: 中國各區域疫情期間和不同季節的流行特徵與關鍵因素, Online.
- 6/6/2020, **Symposium on Atmospheric Environment Monitoring and Simulation & 2020 CIAAS Online Meeting**: Ozone Pollution in China and India: Seasonality and Sources, Online.
- 5/19/2020, **南京信息工程大學環境科學與工程學院**: 氣溶膠生成與天氣、氣候相互作用數值模擬, Online.
- 4/5/2020, **EGU General Assembly 2020**: Aerosol Pollution in Asia and its Interactions with Climate, Online.
- 11/10/2019, **Workshop of State Key Joint Laboratory of Environment Simulation and Pollution Control, Beijing**: China's Clean Air Action Has Suppressed Unfavorable Influences of Climate on Wintertime $PM_{2.5}$ in Beijing, Beijing, China.
- 10/17/2019, **Chinese University of Hong Kong Geography and Resource Management Seminar Series**: Air Pollution in Asia and its Interactions with Health and Climate, Hong Kong.
- 05/31/2019, **Hong Kong Polytechnic University Department of Civil and Environmental Engineering Seminar**: Feeling the Pulse of the Ocean, Hong Kong.
- 04/11/2019, **Max Planck Institute for Chemistry Seminar**: Seasonal Prediction of Indian Wintertime Aerosol Pollution using the Ocean Memory Effect, Mainz, Germany.
- 03/05/2019, **University of Science and Technology of China Seminar**: Seasonal Prediction of Indian Wintertime Aerosol Pollution using the Ocean Memory Effect, Hefei, China.
- 10/12/2018, **Nanjing University of Information Science and Technology Atmospheric Physics Seminar**: Secular Decrease of Wind Power Potential in India Associated with Warming Indian Ocean, Nanjing, China.

- 09/27/2018, **Hong Kong Baptist University Department of Geography Public Seminar Series**: Secular Decrease of Wind Power Potential in India Associated with Warming Indian Ocean, Hong Kong.
- 09/26/2018, **City University of Hong Kong School of Energy and Environment Seminar**: Secular Decrease of Wind Power Potential in India Associated with Warming Indian Ocean, Kowloon, Hong Kong.
- 03/18/2018, **Peking University Excellent Youth Talent International Forum**: Combining Atmospheric Modeling and Observations to Study Aerosol Pollution in North China, Beijing, China.
- 07/30/2017, **Atmospheric Chemistry Colloquium for Emerging Senior Scientists (ACCESS XIV)**: Improving the Understanding of Haze Pollution in the North China Plain via Atmospheric Modeling and Data Assimilation, New York, USA.
- 04/17/2017, **Tsinghua University Earth System Sciences Seminar**: Combining Atmospheric Modeling and Observations to Study Aerosol Pollution in North China, Beijing, China.
- 04/12/2017, **State Key Laboratory of Atmospheric Boundary Layer Physics and Atmospheric Chemistry (LAPC), Chinese Academy of Sciences Seminar**: Combining Atmospheric Modeling and Observations to Study Aerosol Pollution in North China, Beijing, China.
- 03/28/2017, **Nanjing University School of Environment Seminar**: Combining Atmospheric Modeling and Observations to Study Aerosol Pollution in North China, Nanjing, China.
- 03/23/2017, **Sun Yat-Sen University Atmospheric Sciences Seminar**: Combining Atmospheric Modeling and Observations to Study Aerosol Pollution in North China, Guangzhou, China.
- 03/22/2017, **State Key Laboratory of Organic Geochemistry (SKLOG), Chinese Academy of Sciences Seminar**: Haze Pollution in Northern China from a Modeling Perspective, Guangzhou, China.
- 01/20/2017, **Yale University**: Combining Atmospheric Modeling and Observations to Study Aerosol Pollution in North China, New Haven, CT, USA.
- 05/02/2016, **Harvard University China Project Seminar**: Haze Pollution in Northern China from a Modeling Perspective, Cambridge, MA, USA.
- 03/22/2016, **Yonsei University Atmospheric Sciences Seminar**: Improving the Understanding of Haze Pollution in the North China Plain via Atmospheric Modeling and Data Assimilation, Seoul, South Korea.
- 03/18/2016, **Nanjing University of Information Science and Technology Atmospheric Physics Seminar**: Haze Pollution in the North China Plain, Nanjing, China.
- 03/01/2016, **International Space Science Institute Workshop on Persistent Regional Air Pollution in Asia**: Chemical and Meteorological Feedbacks in the Formation of Intense Haze Events, Beijing, China.
- 07/06/2015, **World Meteorological Organization (WMO) Near Real Time (NRT) Data Application Pilot Project Workshop**: Modeling Study of the 2010 Regional Haze Event in the North China Plain, Beijing, China.

## Media Coverage

- Brand Hong Kong: Dr. Gao Meng is named in the Forbes 30 Under 30 Asia Class of 2021: https://www.forbes.com/profile/gao-meng/?list=30under30-asia-healthcare-science&sh=556654282046&fbclid=IwAR1yPxc2ihUtXpxERbY_hKWGDuiff9JnMGyXBu7oD_oAmSdPTHO-QSKW1Bw
- Care for the air: https://bunews.hkbu.edu.hk/news/people/care-for-the-air
- Dr Gao Meng named in Forbes China 30 Under 30 List: https://bunews.hkbu.edu.hk/news/people/dr-gao-meng-named-in-forbes-china-30-under-30-list
- Atmospheric scientist joins global meteorological committee: https://bunews.hkbu.edu.hk/news/global/atmospheric-scientist-joins-global-meteorological-committee
- HKBU scholar receives prestigious award from European Geosciences Union for achievements in atmospheric sciences: https://cpro.hkbu.edu.hk/en/press_release/detail/HKBU-scholar-receives-prestigious-award-from-European-Geosciences-Union-for-achievements-in-atmospheric-sciences/

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- New computer model can predict India's pollution levels months earlier
  *DNA India*: https://www.dnaindia.com/india/report-new-computer-model-can-predict-india-s-pollution-levels-months-earlier-2777336
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  *The Indian Express*: https://indianexpress.com/article/technology/tech-news-technology/new-computer-model-can-predict-indias-pollution-levels-months-earlier-5863129/
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- Predicting pollution levels using oceans' memory:
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- El Niño may trigger spike in pollution: study findings to aid plan to control emissions:
  *The Telegraph*: https://www.telegraphindia.com/india/el-nino-may-trigger-spike-in-pollution-study/cid/1694692
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  *PHYS.ORG*: https://phys.org/news/2019-07-el-nio-antarctic-oscillation-air.html
- India's wind power potential decreasing with climate change:
  *New Kerala*: https://www.newkerala.com/news/read/76831/indias-wind-power-potential-decreasing-with-climate-change.html
  *ANI News*: https://www.aninews.in/news/indias-wind-power-potential-decreasing-with-climate-change201812091628070001/
  *The Siasat Daily*: https://archive.siasat.com/news/indias-wind-power-potential-decreasing-climate-change-1443459/
  *Business Standard*: https://www.business-standard.com/article/news-ani/india-s-wind-power-potential-decreasing-with-climate-change-118120900317_1.html
- Wind power vulnerable to climate change in India:
  *EurekAlert*: https://www.eurekalert.org/pub_releases/2018-12/hjap-wpv120518.php
  *Tech Xplore*: https://techxplore.com/news/2018-12-power-vulnerable-climate-india.html
  *Harvard News*: https://www.seas.harvard.edu/news/2018/12/wind-power-vulnerable-climate-change-india
  *Science Daily*: https://www.sciencedaily.com/releases/2018/12/181205142703.htm
- Climate change may hit India's wind power:
  *The Hindu*: https://www.thehindu.com/sci-tech/science/climate-change-may-hit-indias-wind-power/article26688356.ece
- Study shows climate change impacts wind energy industry:
  *Mongabay India*: https://india.mongabay.com/2019/03/study-shows-climate-change-impacts-wind-energy-industry/
- Climate change affecting India's wind power generation:
  *Saur Energy International*: https://www.saurenergy.com/solar-energy-news/climate-change-affecting-indias-wind-power-generation
- India's dropping wind-power potential:
  *Harvard Magazine*: https://harvardmagazine.com/2018/12/india-windpower-decline
- India's wind power potential declining due to global warming:
  *Energy World*: https://energy.economictimes.indiatimes.com/news/renewable/indias-wind-power-potential-declining-due-to-warming-study/66967058
- China's Action on Air Quality is Saving Lives:
  *Climate News Network*: https://climatenewsnetwork.net/air-quality/
  *Eco-Business*: https://www.eco-business.com/news/chinas-action-on-air-quality-is-saving-lives/
- Emissions from power plants responsible for premature deaths:
  *Research Matters*: https://researchmatters.in/news/emissions-power-plants-responsible-premature-

deaths-finds-study

- Eliminating Emissions in India and China Could Add Years to People's Lives:
  *Science Daily*: https://www.sciencedaily.com/releases/2018/10/181017172832.htm
  *PHYS.ORG*: https://phys.org/news/2018-10-emissions-india-china-years-people.html
- Phantom particles: estimate claims $PM_{2.5}$ air restrictions in China will mean virtual lives saved:
  *American Council on Science and Health*: https://www.acsh.org/news/2018/10/24/phantom-particles-estimate-claims-pm25-air-restrictions-china-will-mean-virtual-lives-saved-13531
- Cleaner power sources could save 26 million years of life:
  *Asian Scientist*: https://www.asianscientist.com/2018/10/in-the-lab/india-china-pollution-health/

## Service

- **Journal Reviewer**: *Nature Energy, Nature Communications, PNAS*, etc.
- **Proposal Reviewer:** *Harvard CHAN-NIEHS Center for Environmental Health Pilot Project*
- **Conferences Session Convener and Chair:** *AGU, AOGS, etc.*